Exp. No: 2 Electric Field and Electric Flux Density in a Sphere Full of Charges with Varying Radius

Aim:

To write and validate MATLAB programs for electric field intensity (E) and electric flux density (D) for specified spherical charge distributions.

Objectives:

1. To derive expressions for electric field intensity (E) and electric flux density (D) inside and outside a uniformly charged sphere using Gauss's law.

2. To plot the variation of E and D with radial distance for varying sphere radius and charge density.

3. To validate the simulated results against theoretical calculations for different radii.

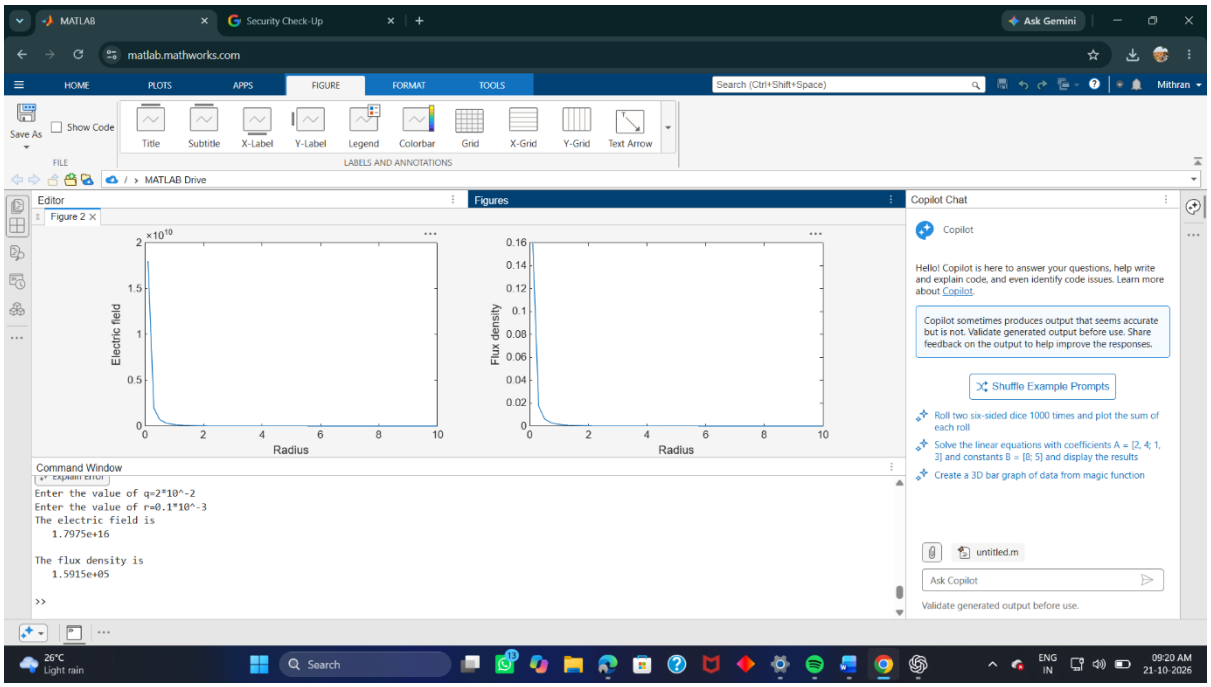
Test case (i) Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius

```
clc;
clear;
q=input('Enter the value of q=');
r=input('Enter the value of r=');
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
E=k*q/(r^2);
disp("The electric field is");
disp(E);
D=epsilon*E;
disp("The flux density is"); disp(D);
% Plot x=linspace(0.1,10,50);
E_plot=k*q./(x.^2);
D_plot=epsilon*E_plot; figure;
subplot(1,2,1);
plot(x,E_plot); xlabel('Radius');
```

```
ylabel('Electric field');
subplot(1,2,2);
plot(x,D_plot); xlabel('Radius');
ylabel('Flux density')
```

Output:-

Enter the value of q=2*10^-2 Enter the value of r=0.1*10^-3



Test case (ii) Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0

Program:

```
clc;
clear;
close all;

rho0 = 1e-6;      % Uniform volume charge density (C/m^3)
epsilon0 = 8.854e-12; % Permittivity of free space (F/m)
R = 1;            % Sphere radius (m)
```

```
r = linspace(0, R, 100);

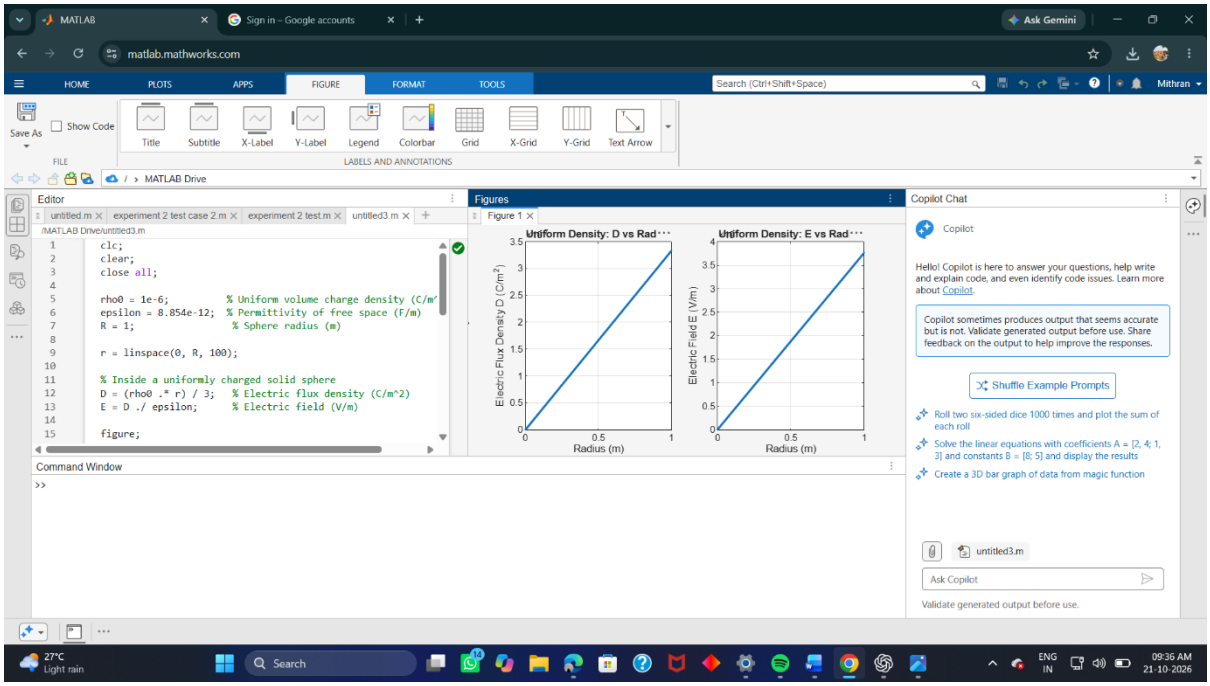

D = (rho0 .* r) ./ 3; % Electric flux density (C/m^2)
E = D ./ epsilon0;    % Electric field (V/m)


subplot(1, 2, 1);
plot(r, D, 'LineWidth', 2);
grid on;
xlabel('Radius (m)');
ylabel('Electric Flux Density D (C/m^2)');
title('Uniform Density: D vs Radius');


subplot(1, 2, 2);
plot(r, E, 'LineWidth', 2);
grid on;
xlabel('Radius (m)');
ylabel('Electric Field E (V/m)');
title('Uniform Density: E vs Radius');
```

Output:

Linear increase in D and Linear increase in E



Test case (iii) For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive and plot the electric field (E) and electric flux density (D)

Program:

```
clc;

clear;

close all;


rho0 = 1e-6;       % Charge density (C/m^3)

epsilon0 = 8.854e-12; % Permittivity of free space (F/m)

R = 1;             % Radius (m)


r = linspace(0, R, 100);

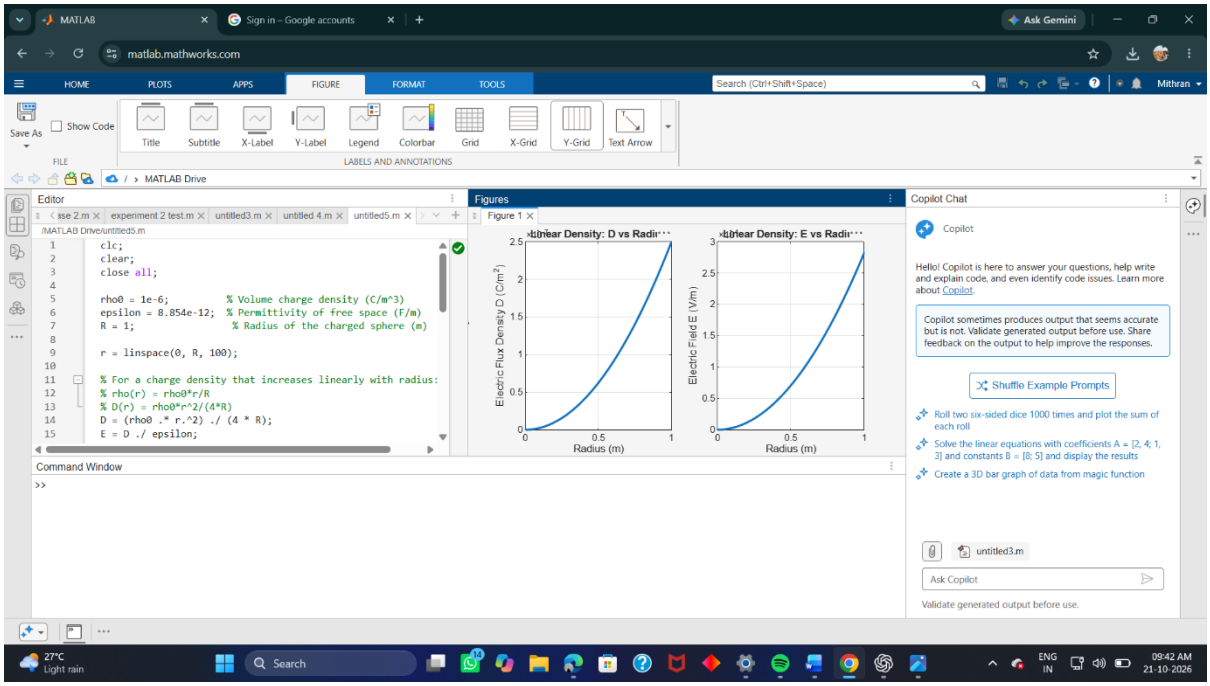

D = (rho0 .* r.^2) ./ 4; % Electric flux density (C/m^2)

E = D ./ epsilon0;      % Electric field (V/m)


subplot(1, 2, 1);
```

```
plot(r, D, 'LineWidth', 2);
grid on;
xlabel('Radius (m)');
ylabel('Electric Flux Density D (C/m^2)');
title('Linear Density: D vs Radius');

subplot(1, 2, 2);
plot(r, E, 'LineWidth', 2);
grid on;
xlabel('Radius (m)');
ylabel('Electric Field E (V/m)');
title('Linear Density: E vs Radius');
```



Test case (iv): For a uniformly charged solid sphere of radius R = 2 m and volume charge density $\rho_v = 3\times10^{-6}$ C/m^3, calculate and plot E and D inside and outside the sphere.

Program:

```
clc;
clear;
```

```
close all;

rho0 = 3e-6;              % Volume charge density (C/m^3)
epsilon0 = 8.854187817e-12;  % Permittivity of free space (F/m)
R = 2;                    % Sphere radius (m)

r = linspace(0.01, 4, 401);  % Radial distance (m); avoid r = 0
E = zeros(size(r));

inside = (r <= R);
outside = (r > R);

E(inside) = rho0 .* r(inside) ./ (3 * epsilon0);
E(outside) = rho0 * R^3 ./ (3 * epsilon0 .* r(outside).^2);

D = epsilon0 .* E;         % Electric flux density (C/m^2)

figure;

subplot(2, 1, 1);
plot(r, E, 'LineWidth', 1.5);
grid on;
xline(R, '--r', 'R');
xlabel('Radius r (m)');
ylabel('Electric Field E (V/m)');
title('Electric Field Inside and Outside a Uniformly Charged Sphere');

subplot(2, 1, 2);
```

```
plot(r, D, 'LineWidth', 1.5);

grid on;

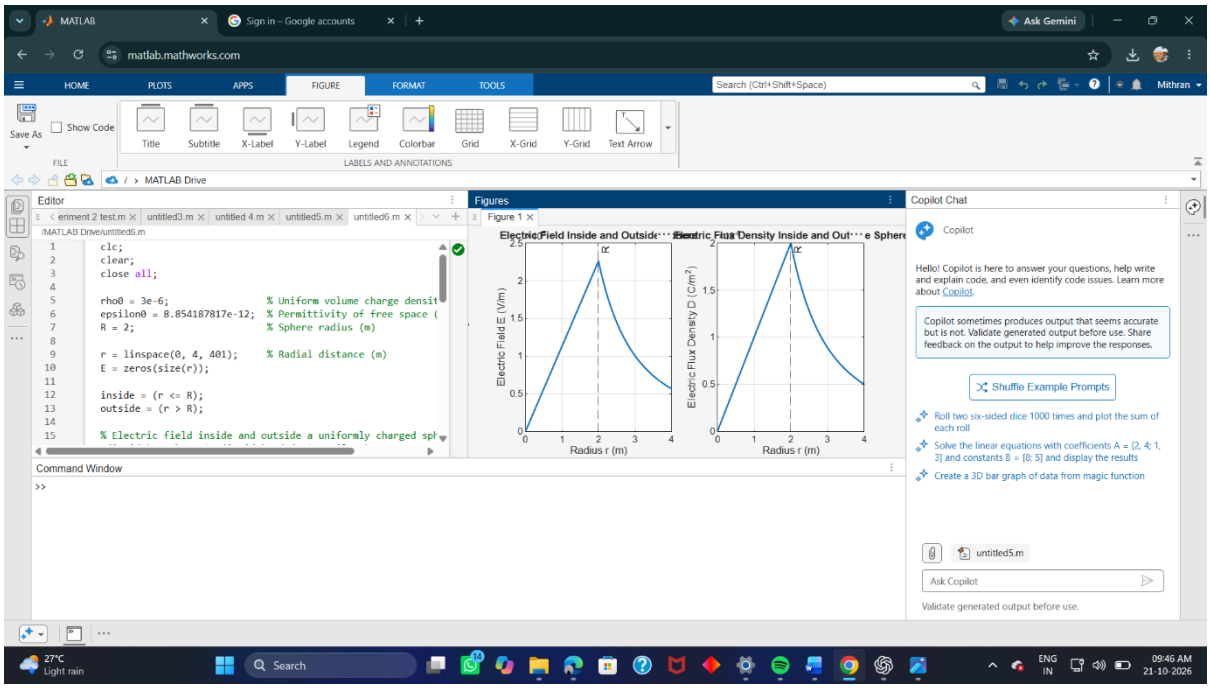
xline(R, '--r', 'R');

xlabel('Radius r (m)');

ylabel('Electric Flux Density D (C/m^2)');

title('Electric Flux Density Inside and

 Outside a Uniformly Charged Sphere');
```



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Select the appropria t e charge distributi o n model	Select the appropriat e charge distributio n model	Correctly distinguishes point, uniform volume and specified non uniform charge models and their assumptions	Selects the correct model with minor clarific a tion needed.	Shows partial understanding of the charge model..	25	2.5		

2	Impleme nt E and D calculatio n s in MATLA B	Implement E and D calculation s in MATLAB	Correctly calculates electric field E and flux density D with consistent SI units.	Calcula t ions are mostly correct with minor coding or unit errors..	Some calculations work but require assistance..	Unable to obtain valid E and D values.	25	2.5	
3	Analyze radial variation and validate result	Analyze radial variation and validate result	Plots the correct inside/outsid e or stated distribution behaviour and verifies against theory	Plots and validati on are mostly correct with minor discrep a ncies.	Plots or theoretical validation are incomplete.	Plots are incorrect or results are not validated.	30	3	
4	Interpret and document findings	nterpret and document findings	Explains the effect of radius and charge distribution and presents labelled plots/tables	Interpre t ation is general l y correct with minor omissio ns.	Interpretatio n is limited or documentati o n incomplete..	Cannot interpret or document results.	20	2	
Tota l							100	10	